

Differentiation - Set 1 - Solutions

Specification MM6.1-MM6.3, MM8.5.

Answer key

Question	Answer	Topic
1	D	Differentiation
2	C	Differentiation
3	F	Differentiation
4	B	Differentiation
5	F	Differentiation
6	B	Differentiation
7	A	Differentiation
8	E	Differentiation
9	C	Differentiation
10	A	Differentiation
11	D	Differentiation
12	B	Differentiation

Worked solutions

Question 1 DIF-01 · Differentiation · spec MM6.2

Find $\frac{dy}{dx}$ when $y = \frac{(3x-2)^2}{x^2}$.

Answer: D

$$y = 9 - 12x^{-1} + 4x^{-2}, \text{ so } y' = 12x^{-2} - 8x^{-3}.$$

Full method - how and why

The spec only allows differentiating powers of x , so the expression must first be written as a sum of such powers - expand and divide term by term.

- 1 Expand the numerator: $(3x-2)^2 = 9x^2 - 12x + 4$.

why: There is no quotient or chain rule available, so the fraction must be dismantled.

- 2 Divide each term by x^2 : $y = 9 - 12x^{-1} + 4x^{-2}$.

why: $\frac{9x^2}{x^2} = 9$, $\frac{-12x}{x^2} = -12x^{-1}$, $\frac{4}{x^2} = 4x^{-2}$.

- 3 Differentiate term by term with $\frac{d}{dx}x^n = nx^{n-1}$: $0 + 12x^{-2} - 8x^{-3}$.

why: $-12 \times (-1) = +12$ and $4 \times (-2) = -8$; the constant 9 differentiates to 0.

- 4 Write in fraction form: $\frac{12}{x^2} - \frac{8}{x^3}$.

why: Match the form used in the options.

Where it goes wrong

- Differentiating the bracket directly as $\frac{6(3x-2)}{x^2}$ - that would need the quotient rule (and is wrong anyway).
- Sign slips: differentiating x^{-1} gives $-x^{-2}$, so $-12x^{-1}$ gives $+12x^{-2}$.

Takeaway Rewrite as $\sum ax^n$ before differentiating; negative powers change sign when differentiated.

Question 2 DIF-02 · Differentiation · spec MM6.3

Find the coordinates of the stationary points of $y = x^3 - 3x^2 - 9x + 5$ and state their nature.

Answer: C

$$y' = 3x^2 - 6x - 9 = 3(x-3)(x+1). \quad y'' = 6x - 6: \text{ negative at } x = -1 \text{ (max, } y = -1 - 3 + 9 + 5 = 10\text{), positive at } x = 3 \text{ (min, } y = 27 - 27 - 27 + 5 = -22\text{)}.$$

Full method - how and why

Stationary points from $f' = 0$, their nature from the sign of f'' , and the y -values by substitution.

- 1 Differentiate: $y' = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x-3)(x+1)$.

why: Factor out the 3 first - it makes the factorisation obvious.

- 2 Solve $y' = 0$: $x = 3$ and $x = -1$.

why: Stationary points are where the gradient vanishes.

- 3 Second derivative: $y'' = 6x - 6$. At $x = -1$, $y'' = -12 < 0$ (maximum); at $x = 3$, $y'' = 12 > 0$ (minimum).

why: Negative second derivative means the curve bends downwards - a maximum.

- 4 Find the y -values: $y(-1) = -1 - 3 + 9 + 5 = 10$; $y(3) = 27 - 27 - 27 + 5 = -22$.

why: Substitute into the original function, not the derivative.

Where it goes wrong

- Swapping the natures - for a positive cubic the maximum always comes first (smaller x).
- Substituting into y' instead of y for the coordinates.

Takeaway $f' = 0$ locates stationary points; the sign of f'' classifies them; substitute into f for the heights.

Question 3 DIF-03 · Differentiation · spec MM6.3

The tangent to $y = x^2 - 4x + 1$ at the point where $x = 3$ meets the y-axis at

Answer: F

$y' = 2x - 4 = 2$ at $x = 3$; the point is $(3, -2)$. Tangent: $y + 2 = 2(x - 3)$, so at $x = 0$, $y = -8$.

Full method - how and why

Tangent line: gradient from the derivative, point from the curve, then find where it meets the axis.

- 1 Differentiate: $y' = 2x - 4$; at $x = 3$ the gradient is 2.
why: The derivative evaluated at a point gives the tangent's gradient there.
- 2 Find the point: $y(3) = 9 - 12 + 1 = -2$, so the tangent touches at $(3, -2)$.
why: The tangent passes through the curve at that x - use the original function.
- 3 Write the tangent: $y + 2 = 2(x - 3)$.
why: Point-gradient form with the point and gradient just found.
- 4 Set $x = 0$: $y + 2 = -6$, so $y = -8$. The tangent meets the y-axis at $(0, -8)$.
why: The y-axis is $x = 0$.

Where it goes wrong

- Using $y(3) = -2$ as the intercept and answering $(0, -2)$.
- Sign slip: $y = 2x - 8$, not $2x + 8$.

Takeaway Tangent = gradient from f' + point from f ; then substitute $x = 0$ for the intercept.

Question 4 DIF-04 · Differentiation · spec MM6.3

Find the equation of the normal to the curve $y = \sqrt{x}$ at the point $(4, 2)$.

Answer: B

$y' = \frac{1}{2\sqrt{x}} = \frac{1}{4}$ at $x = 4$, so the normal has gradient -4 : $y - 2 = -4(x - 4)$, i.e. $y = -4x + 18$.

Full method - how and why

Normal line: the negative reciprocal of the tangent's gradient, through the same point.

- 1 Write $y = x^{1/2}$ and differentiate: $y' = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$.
why: Convert the root to a fractional power before differentiating.
- 2 At $x = 4$: $y' = \frac{1}{2 \times 2} = \frac{1}{4}$.
why: $\sqrt{4} = 2$.
- 3 Normal gradient = $-\frac{1}{1/4} = -4$.
why: Normal is perpendicular to the tangent: negative reciprocal.
- 4 Line through $(4, 2)$: $y - 2 = -4(x - 4) \Rightarrow y = -4x + 18$.
why: $-4 \times -4 = +16$, plus the 2.

Where it goes wrong

- Using the tangent gradient $\frac{1}{4}$ and choosing $y = \frac{1}{4}x + 1$.
- Arithmetic on the constant: $y = -4x + 16$ is offered.

Takeaway Normal gradient = $-1/f'(x_1)$; then substitute the point into the point-gradient form.

Question 5 DIF-05 · Differentiation · spec MM6.3

For what range of values of x is $f(x) = 2x^3 - 15x^2 + 24x + 1$ decreasing?

Answer: F

$$f'(x) = 6x^2 - 30x + 24 = 6(x-1)(x-4) < 0 \text{ for } 1 < x < 4.$$

Full method - how and why

'Decreasing' is the condition $f'(x) < 0$ - a quadratic inequality once the derivative is found.

1 Differentiate: $f'(x) = 6x^2 - 30x + 24$.

why: Term by term.

2 Factorise: $6(x^2 - 5x + 4) = 6(x-1)(x-4)$.

why: Taking out the 6 makes the roots visible.

3 Solve $f'(x) < 0$: the upward parabola is negative between its roots, so $1 < x < 4$.

why: The gradient is negative exactly where the function falls.

Where it goes wrong

- Solving $f'(x) > 0$ and answering $x < 1$ or $x > 4$ - which is offered.
- Forgetting the factor 6 when factorising, giving the wrong roots.

Takeaway Decreasing $\Leftrightarrow f' < 0$; find the roots of f' and use the shape of its graph.

Question 6 DIF-06 · Differentiation · spec MM6.1

The displacement s metres of a particle at time t seconds is $s = t^3 - 6t^2 + 9t$. At what times is the particle momentarily at rest?

Answer: B

$$\text{Velocity} = \frac{ds}{dt} = 3t^2 - 12t + 9 = 3(t-1)(t-3) = 0 \text{ at } t = 1 \text{ and } t = 3.$$

Full method - how and why

'At rest' means velocity zero, and velocity is the derivative of displacement.

1 Differentiate the displacement: $v = \frac{ds}{dt} = 3t^2 - 12t + 9$.

why: Velocity is the rate of change of displacement.

2 Factorise: $3(t^2 - 4t + 3) = 3(t-1)(t-3)$.

why: Take the 3 out first.

3 Set $v = 0$: $t = 1$ and $t = 3$.

why: 'Momentarily at rest' is exactly $v = 0$.

Where it goes wrong

- Setting $s = 0$ instead of $v = 0$, which gives $t = 0$ and $t = 3$ - an offered option.
- Forgetting one of the two roots.

Takeaway Displacement $\rightarrow$ velocity $\rightarrow$ acceleration by differentiating; 'at rest' means the velocity is zero, not the displacement.

Question 7 DIF-07 · Differentiation · spec MM6.3

The curve $y = ax^2 + bx$ has a stationary point at $(2, -8)$. Find $a + b$.

Answer: A

$$y' = 2ax + b = 0 \text{ at } x = 2: b = -4a. y(2) = 4a + 2b = 4a - 8a = -4a = -8 \Rightarrow a = 2, b = -8. a + b = -6.$$

Full method - how and why

A stationary point gives two conditions: the derivative vanishes there, and the point is on the curve.

- 1 Differentiate: $y' = 2ax + b$. At $x = 2$: $4a + b = 0$, so $b = -4a$.
why: Stationary means the gradient is zero at that x .
- 2 The point is on the curve: $y(2) = 4a + 2b = -8$.
why: Substitute $x = 2$ into the original function and set it equal to the given y .
- 3 Substitute $b = -4a$: $4a - 8a = -4a = -8$, so $a = 2$ and $b = -8$.
why: One equation in one unknown after substitution.
- 4 Compute $a + b = 2 + (-8) = -6$.
why: The question asks for the sum.

Where it goes wrong

- Using only one of the two conditions.
- Sign error giving $a = -2$ and the answer 6.

Takeaway 'Stationary point at (p, q) ' is two equations: $f'(p) = 0$ and $f(p) = q$.

Question 8 DIF-09 · Differentiation · spec MM6.3

A rectangle has perimeter 40. What is its maximum possible area?

Answer: E

$$\text{Sides } x \text{ and } 20 - x: A = 20x - x^2, A' = 20 - 2x = 0 \text{ at } x = 10, \text{ giving } A = 100 \text{ (a square).}$$

Full method - how and why

A standard optimisation: express the area in one variable using the constraint, then differentiate.

- 1 Let one side be x ; the perimeter condition $2x + 2y = 40$ gives $y = 20 - x$.
why: Use the constraint to eliminate the second variable - optimisation needs a single variable.
- 2 Write the area: $A = x(20 - x) = 20x - x^2$.
why: A quadratic in x .
- 3 Differentiate and solve: $A' = 20 - 2x = 0$ at $x = 10$.
why: Or complete the square: $A = 100 - (x - 10)^2$, which shows the maximum immediately.
- 4 Compute the area: $A = 10 \times 10 = 100$ (a square).
why: $A'' = -2 < 0$ confirms a maximum.

Where it goes wrong

- Using $x + y = 40$ instead of $2x + 2y = 40$, giving 400 - an offered option.
- Reporting the side length 10 rather than the area.

Takeaway For fixed perimeter, the rectangle of greatest area is the square - but show it via the constraint and the derivative.

Question 9 DIF-10 · Differentiation · spec MM6.3

The curve $y = x^3 + px^2 + qx$ has stationary points at $x = 1$ and $x = 3$. Find q .

Answer: C

$$y' = 3x^2 + 2px + q = 3(x-1)(x-3) = 3x^2 - 12x + 9. \text{ So } q = 9 \text{ (and } p = -6\text{).}$$

Full method - how and why

Known stationary points mean known roots of f' . Compare the factorised derivative with its expanded form.

- 1 Differentiate: $y' = 3x^2 + 2px + q$.
why: A cubic's derivative is a quadratic.
- 2 The stationary points are its roots, so $y' = 3(x-1)(x-3)$.
why: The leading coefficient must be 3 to match $3x^2$.
- 3 Expand: $3(x^2 - 4x + 3) = 3x^2 - 12x + 9$.
why: Multiply out to compare coefficients.
- 4 Compare constants: $q = 9$ (and comparing x terms gives $2p = -12$, so $p = -6$).
why: The question asks only for q .

Where it goes wrong

- Forgetting the leading 3 and getting $q = 3$.
- Answering -6 , which is p .

Takeaway Stationary points are roots of f' : build f' in factorised form with the correct leading coefficient.

Question 10 DIF-12 · Differentiation · spec MM6.3

The line $y = 4x + c$ is a tangent to $y = x^2 + 2x + 5$. Find c .

Answer: A

$$2x + 2 = 4 \Rightarrow x = 1, y = 8. 8 = 4 + c \Rightarrow c = 4.$$

Full method - how and why

Tangency via calculus: the gradients must match, which locates the point of contact.

- 1 Differentiate the curve: $y' = 2x + 2$.
why: The tangent's gradient equals the curve's gradient at the contact point.
- 2 Set it equal to the line's gradient: $2x + 2 = 4$, so $x = 1$.
why: This finds **where** the tangent touches.
- 3 Find the y -value on the curve: $y(1) = 1 + 2 + 5 = 8$.
why: The contact point is on the curve.
- 4 The point must also be on the line: $8 = 4(1) + c$, so $c = 4$.
why: Substituting the contact point into the line's equation gives c .

Where it goes wrong

- Using the discriminant route but mis-expanding and getting $c = 1$ or 5 (both offered).
- Substituting into the line the y -value from the wrong x .

Takeaway Tangency: equal gradients give the x ; equal y -values give the constant.

Question 11 DIF-14 · Differentiation · spec MM6.2

Find $\frac{d^2y}{dx^2}$ when $y = \frac{1}{x^2} + 4\sqrt{x}$.

Answer: D

$$y' = -2x^{-3} + 2x^{-1/2}; y'' = 6x^{-4} - x^{-3/2}.$$

Full method - how and why

Two differentiations of powers, with negative and fractional indices. Rewrite first.

1 Rewrite: $y = x^{-2} + 4x^{1/2}$.

why: Roots and reciprocals must become powers before the rule applies.

2 First derivative: $y' = -2x^{-3} + 4 \times \frac{1}{2}x^{-1/2} = -2x^{-3} + 2x^{-1/2}$.

why: $-2 \times x^{-2-1}$ and $\frac{1}{2} \times 4 = 2$.

3 Second derivative: $y'' = -2(-3)x^{-4} + 2(-\frac{1}{2})x^{-3/2} = 6x^{-4} - x^{-3/2}$.

why: Differentiate again, watching the signs: -3 times -2 gives $+6$.

4 Write it as $\frac{6}{x^4} - x^{-3/2}$.

why: Matching the option format.

Where it goes wrong

- Losing a sign and answering $\frac{6}{x^4} + x^{-3/2}$.
- Stopping after the first derivative.

Takeaway Differentiate twice, carefully: each negative index gains one more negative factor.

Question 12 DIF-17 · Differentiation · spec MM6.3

The curve $y = x^2 + kx$ (with $k < 0$) crosses the x -axis at the origin and at one other point P . The gradient of the curve at P is 6. Find k .

Answer: B

$$P = (-k, 0). y' = 2x + k = -2k + k = -k = 6, \text{ so } k = -6.$$

Full method - how and why

Find the second crossing point in terms of k , then impose the gradient condition there.

1 Find the x -intercepts: $x^2 + kx = x(x + k) = 0$, so $x = 0$ or $x = -k$. Thus $P = (-k, 0)$.

why: With $k < 0$, $-k$ is positive - the curve crosses to the right of the origin.

2 Differentiate: $y' = 2x + k$.

why: The gradient at any point.

3 Evaluate at P : $y'(-k) = -2k + k = -k$.

why: Substituting $x = -k$.

4 Set equal to 6: $-k = 6$, so $k = -6$.

why: Consistent with the given condition $k < 0$ ✓.

Where it goes wrong

- Using $x = k$ for the second intercept, giving $k = 2$ or $3k = 6$.
- Answering 6 and ignoring the stated sign of k .

Takeaway Find the point in terms of the parameter first, then apply the gradient condition; check any sign restriction at the end.